

Week 1 Quiz - Recurrent Neural Networks

1. Suppose your training examples are sentences (sequences of words). Which of the following refers to the j th word in the i th training example?

◦ $x^{(i)}_{<j>}$

We index into the i -th row first to get the i th training example (represented by parentheses), then the j -th column to get the j th word (represented by the brackets).

2. Consider this RNN: This specific type of architecture is appropriate when:

◦ $T_x = T_y$

3. To which of these tasks would you apply a many-to-one RNN architecture? (Check all that apply).

- Sentiment classification (input a piece of text and output a 0/1 to denote positive or negative sentiment)
- Gender recognition from speech (input an audio clip and output a label indicating the speaker's gender)

4. At the t -th time step, what is the RNN doing? Choose the best answer.

◦ Estimating $P(y^{<t>} \mid y^{<1>}, y^{<2>}, \dots, y^{<t-1>})$

5. You have finished training a language model RNN and are using it to sample random sentences, as follows: What are you doing at each time step t ?

- (i) Use the probabilities output by the RNN to randomly sample a chosen word for that time-step as $y^{<t>}$. (ii) Then pass this selected word to the next time-step.

6. You are training an RNN, and find that your weights and activations are all taking on the value of NaN ("Not a Number"). Which of these is the most likely cause of this problem?

◦ Exploding gradient problem.

7. Suppose you are training a LSTM. You have a 10000 word vocabulary, and are using an LSTM with 100-dimensional activations a . What is the dimension of Γ_u at each time step?

◦ 100

8. Here're the update equations for the GRU. Alice proposes to simplify the GRU by always removing the Γ_u . I.e., setting $\Gamma_u = 1$. Betty proposes to simplify the GRU by removing the Γ_r . I. e., setting $\Gamma_r = 1$ always. Which of these models is more likely to work without vanishing gradient problems even when trained on very long input sequences?

- Betty's model (removing Γ_r), because if $\Gamma_u \approx 0$ for a timestep, the gradient can propagate back through that timestep without much decay.

9. Here are the equations for the GRU and the LSTM: From these, we can see that the Update Gate and Forget Gate in the LSTM play a role similar to _____ and _____ in the GRU. What should go in the the blanks?

- Γu and $1 - \Gamma u$

10. You have a pet dog whose mood is heavily dependent on the current and past few days' weather. You've collected data for the past 365 days on the weather, which you represent as a sequence as $x_{<1>}, \dots, x_{<365>}$. You've also collected data on your dog's mood, which you represent as $y_{<1>}, \dots, y_{<365>}$. You'd like to build a model to map from $x \rightarrow y$. Should you use a Unidirectional RNN or Bidirectional RNN for this problem?

- Unidirectional RNN, because the value of y depends only on $x_{<1>}, \dots, x_t$, but not on $x_{t+1}, \dots, x_{<365>}$